

Lecture 4: Inference for Subpopulation Proportions and Differences

September 3, 2023

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- ▶ We saw that for large enough sample sizes, our sample proportion/mean was likely to be close to the population proportion/mean.
- ▶ We also demonstrated that we could predict how close we might be by using confidence intervals.
- ▶ We explored a bootstrapping approach and a plug-in approach to producing confidence intervals.

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- ▶ Returning to the turnout example, we will consider Black turnout and the difference between Black and non-Black turnout in the 2020 Georgia presidential election.
- ▶ Even when the overall sample size is fixed (625 for our example), different random samples will produce different subgroup sample sizes, and this adds a layer of complexity.

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- ▶ Returning to the turnout example, we will consider Black turnout and the difference between Black and non-Black turnout in the 2020 Georgia presidential election.
- ▶ Even when the overall sample size is fixed (625 for our example), different random samples will produce different subgroup sample sizes, and this adds a layer of complexity.
- ▶ This complexity will not complicate our analysis much but it does complicate the justification for the analysis.

2020 Presidential Election Turnout in Georgia

We return to the turnout application, but we are now interested in Black turnout and Black turnout in comparison to non-Black turnout. Recall that ...

- ▶ based on a survey of 625 registered voters a week before the election, we got a turnout rate of .68,
- ▶ after the election, we found that the turnout rate among all registered voters was .70

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- ▶ in our survey of 625 registered voters, 181 were Black with a turnout rate of .73 and 444 were non-Black with a turnout rate of .66.

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Suppose we also want to predict Black turnout and the difference between Black and non-Black turnout.

- ▶ in our survey of 625 registered voters, 181 were Black with a turnout rate of .73 and 444 were non-Black with a turnout rate of .66.
- ▶ after the election, we found that the turnout rate among the 30% of registered voters who were black was .74 and among the 70% of non-Black registered voters was .68.

Four Questions

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3. The survey turnout gap of .07 is close to the population turnout gap of .06. Did we get lucky?
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Sampling Distribution for Black Turnout Proportion

Let the variable X take the value 1 if Black and 0 if non-Black

i	1	2	...	625		
	Y_1, X_1	Y_2, X_2	...	Y_{625}, X_{625}	$\bar{Y}_{625}$	$\frac{\sum_{i=1}^{625} Y_i X_i}{\sum_{i=1}^{625} X_i}$
	1,0	1,1	...	0,0	.68	.73

- ▶ Note that $\sum_{i=1}^{625} Y_i X_i$ is not a binomial RV because different samples have different numbers of Black and non-Black individuals.
- ▶ To simulate the sampling distribution (or bootstrapped sampling distribution), we repeatedly sample with replacement individuals and their (Y_i, X_i) variable pairs from the population (or the sample).

Sampling Distribution for Black Turnout Proportion

Let the variable X take the value 1 if Black and 0 if non-Black

i	1	2	...	625		
s	Y_1, X_1	Y_2, X_2	...	Y_{625}, X_{625}	$\bar{Y}_{625}$	$\frac{\sum_{i=1}^{625} Y_i X_i}{\sum_{i=1}^{625} X_i}$
1	1,0	1,1	...	0,0	.68	.73
2	0,0	1,0	...	1,1	.70	.72

- ▶ Note that $\sum_{i=1}^{625} Y_i X_i$ is not a binomial RV because different samples have different numbers of Black and non-Black individuals.
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Sampling Distribution for Black Turnout Proportion

Let the variable X take the value 1 if Black and 0 if non-Black

i	1	2	...	625		
s	Y_1, X_1	Y_2, X_2	...	Y_{625}, X_{625}	$\bar{Y}_{625}$	$\frac{\sum_{i=1}^{625} Y_i X_i}{\sum_{i=1}^{625} X_i}$
1	1,0	1,1	...	0,0	.68	.73
2	0,0	1,0	...	1,1	.70	.72
3	1,0	1,1	...	0,1	.75	.76
4	1,1	1,0	...	0,0	.73	.73
5	0,0	0,0	...	1,1	.69	.71
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	1,0	0,1	...	1,1	.74	.75
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

- ▶ Note that $\sum_{i=1}^{625} Y_i X_i$ is not a binomial RV because different samples have different numbers of Black and non-Black individuals.
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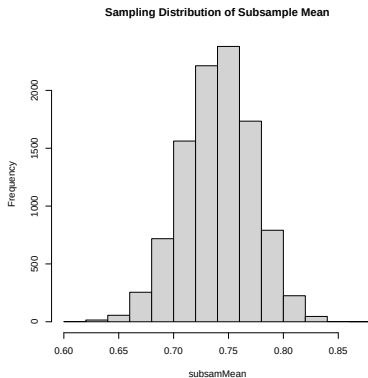
Simulating the Sampling Distribution

<i>ID</i>	1	2	3	4	...	...	...	...	7,233,584	Mean
<i>y_{ID}</i>	0	1	0	1	...	...	...	...	1	.70
<i>x_{ID}</i>	0	0	1	0	...	...	...	...	1	.30

For each row of the sampling distribution,

```
dfprop = data.frame(y,x)
ind = sample(1:7233584, size=625, replace=TRUE)
dfind = df[ind,]
subsamMean = mean(dfind$y[dfind$x == 1])
# alternative approach
#subsamMean = sum(dfind$y*dfind$x)/sum(dfind$x)
```

```
S = 10000
subsamMean = rep(NA,S)
for( s in 1:S ){
  ind = sample(1:N,size=625,replace=T)
  dfind = dfpop[ind,]
  subsamMean[s] = mean(dfind$y[dfind$x==1])
}
```



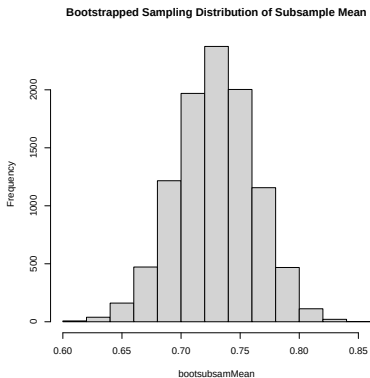
Simulating the Bootstrapped Sampling Distribution

i	1	2	...	625	$\bar{y}_{625}$
ID	5093883	5473235	...	3042073	
y_i	1	1	...	0	.68
x_i	1	0	...	0	.29

Starting with the sample (first row of sampling distribution), each row of the bootstrapped sampling distribution is generated as the following:

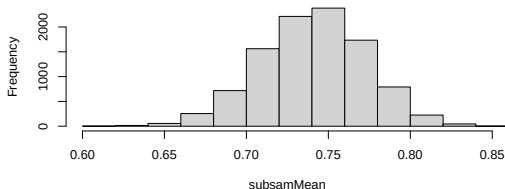
```
bootind = sample(1:625, size=625, replace=TRUE)
dfboot = dfsam[bootind,]
subsamMean = mean(dfboot$y[dfboot$x == 1])
# alternative approach
#subsamMean = sum(dfboot$y*dfboot$x)/sum(dfboot$x)
```

```
B = 10000  
bootsubsamMean = rep(NA,B)  
for( b in 1:B ){  
  bootind = sample(1:625,size=625,replace=T)  
  dfboot = dfsam[bootind,]  
  bootsubsamMean[b] = mean(dfboot$y[dfboot$x==1])  
}
```

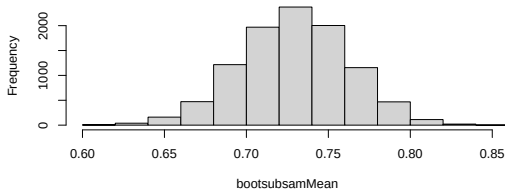


Sampling Distribution vs Bootstrapped Distribution

Sampling Distribution of Subsample Mean



Bootstrapped Sampling Distribution of Subsample Mean



```
> quantile(bootsubsamMean,prob =c(.025,.975))  
      2.5%      97.5%  
0.6632653 0.7926845
```

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2. Could we have predicted how close we would get to .74 before the election? Yes, using bootstrapped CI.

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2. Could we have predicted how close we would get to .74 before the election? Yes, using bootstrapped CI.
3. The survey turnout gap of .07 is close to the population turnout gap of .06. Did we get lucky?
4. Could we have predicted how close we would get to a .06 turnout gap before the election?

Black Turnout Minus non-Black Turnout

i	1	2	...	625		
	Y_1, X_1	Y_2, X_2	...	Y_{625}, X_{625}	$\frac{\sum_{i=1}^{625} Y_i X_i}{\sum_{i=1}^{625} X_i}$	$\frac{\sum_{i=1}^{625} Y_i X_i}{\sum_{i=1}^{625} X_i} - \frac{\sum_{i=1}^{625} Y_i (1 - X_i)}{\sum_{i=1}^{625} (1 - X_i)}$
	1,0	1,1	...	0,0	.73	.07

- ▶ Note that $\sum_{i=1}^{625} Y_i X_i$ is not a binomial RV because different samples have different numbers of Black and non-Black individuals.
- ▶ To simulate the sampling distribution (or bootstrapped sampling distribution), we repeatedly sample with replacement individuals and their (Y_i, X_i) variable pairs from the population (or the sample).

Black Turnout Minus non-Black Turnout

i	1	2	...	625		
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1	1,0	1,1	...	0,0	.73	.07
2	0,0	1,0	...	1,1	.70	.03

- Note that $\sum_{i=1}^{625} Y_i X_i$ is not a binomial RV because different samples have different numbers of Black and non-Black individuals.
- To simulate the sampling distribution (or bootstrapped sampling distribution), we repeatedly sample with replacement individuals and their (Y_i, X_i) variable pairs from the population (or the sample).

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1	1,0	1,1	...	0,0	.73	.07
2	0,0	1,0	...	1,1	.70	.03
3	1,0	1,1	...	0,1	.76	.09
4	1,1	1,0	...	0,0	.73	.06
5	0,0	0,0	...	1,1	.71	.04
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	1,0	0,1	...	1,1	.74	.07
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

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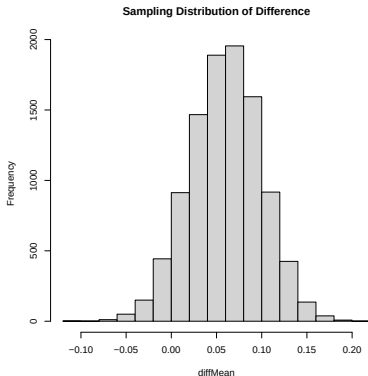
Simulating the Sampling Distribution of Differences

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<i>y_{ID}</i>	0	1	0	1	...	...	...	...	1	.70
<i>x_{ID}</i>	0	0	1	0	...	...	...	...	1	.30

For each row of the sampling distribution,

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dfprop = data.frame(y,x)
ind = sample(1:7233584, size=625, replace=TRUE)
dfind = df[ind,]
diffMean = mean(dfind$y[dfind$x == 1]) -
mean(dfind$y[dfind$x == 0])
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diffMean = rep(NA,S)
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  diffMean[s] = mean(dfind$y[dfind$x==1]) -
  mean(dfind$y[dfind$x==0])
}
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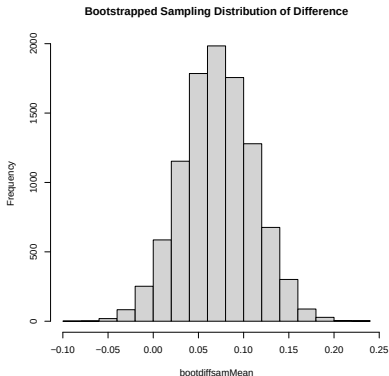
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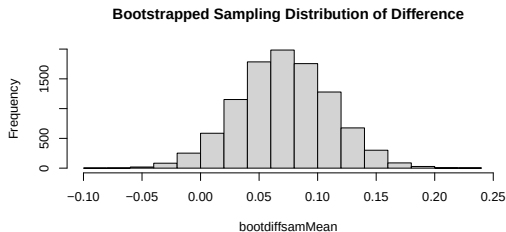
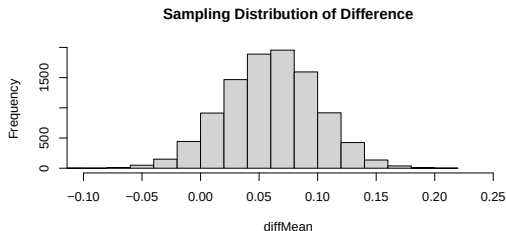
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```
> quantile(bootdiffsamMean,prob =c(.025,.975))  
      2.5%      97.5%  
-0.006698302  0.148697667
```

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1. .73 is close to the Black turnout rate of .74. Did we get lucky? Maybe a little, sampling distribution from .65 to .85
2. Could we have predicted how close we would get to .74 before the election? Yes, using bootstrapped CI.

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2. Could we have predicted how close we would get to .74 before the election? Yes, using bootstrapped CI.
3. The survey turnout gap of .07 is close to the population turnout gap of .06. Did we get lucky? A bit, sampling distribution can go from 0 to .15
4. Could we have predicted how close we would get to a .06 turnout gap before the election? Yes, CI gives correct sense of uncertainty.

Summary

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